Basic Buck-Boost Converter

Objective

This note examines the open-loop behavior of a non-isolated buck-boost converter using PSIM simulation.
The focus is on:

- The relationship between duty cycle and output voltage magnitude.
- The effect of inductance and capacitance on transient response in both step-down and step-up regions.

The goal is to observe how identical passive components behave under both operating modes before introducing closed-loop control.

System Model

Input voltage:25VDC
Load:5Ω
Control method: Open-loop PWM
Duty ratios evaluated: D=0.3 (buck region), D= 0.7 (boost region)
Switching frequency:20kHz

Operating Principle

When the MOSFET switch is **ON**, current flows from the input through the inductor, storing energy in its magnetic field:

$$V_{L,on} = V_{in}$$

When the switch turns **OFF**, the inductor reverses polarity and releases energy through the diode into the output capacitor and load:

$$V_{L,off} = V_{in} + V_{out}$$

The steady-state voltage relationship is given by:

$$|V_{out}| = \frac{D}{1-D} V_{in}$$

where is the D duty cycle.

Circuit Parameters/Simulation Conditions

PARAMETER	SYMBOL	VALUE	DESCRIPTION
INPUT VOLTAGE	(Vin)	25 V	Constant DC source
LOAD RESISTANCE	(R)	5 Ω	Represents load
INDUCTANCE	(L)	5mH	Controls current ramp
CAPACITANCE	(C)	200 µF	Smooths voltage ripple
SWITCHING FREQUENCY	(fs)	20 kHz	Determines ripple and response
DUTY CYCLE	(D)	0.3 / 0.7	Buck and boost test points

These parameters ensured continuous conduction and allowed both buck and boost behavior to be observed under identical hardware conditions.

Circuit Diagram

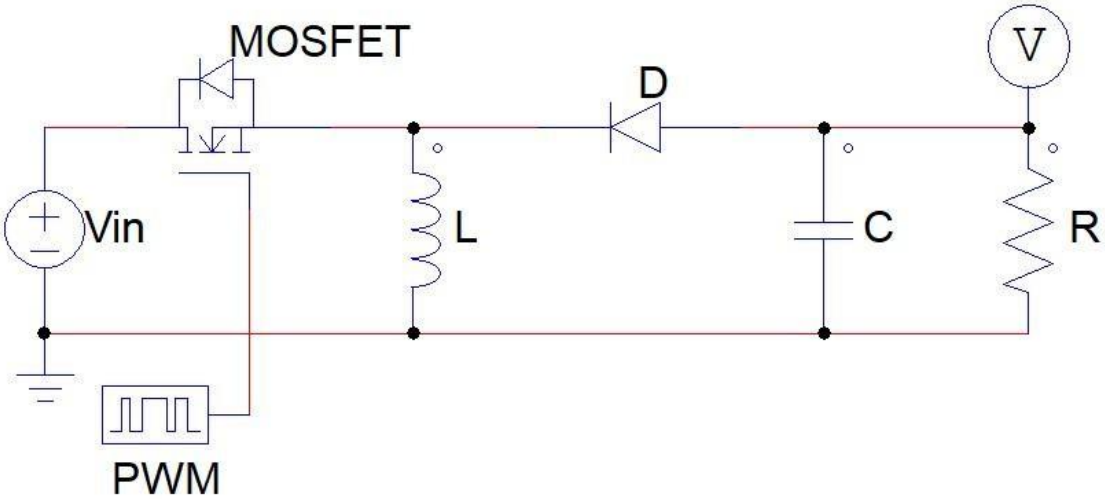


Figure 1

Simulation Results and Analysis

(a) Step-Down Region (D = 0.3)

- Output stabilized near −11 V, consistent with theory.
- A short startup overshoot was observed before settling.
- Average inductor current was approximately 3 A.

The overshoot is attributed to stored magnetic energy continuing to charge the output capacitor during the initial transient.

(b) Step-Up Region (D = 0.7)

- Output reached approximately −58 V, consistent with theoretical prediction.
- The voltage rise was smooth and monotonic.
- Average inductor current increased significantly (approximately 38 A).

Higher duty operation required substantially larger inductor current to maintain power balance between input and output.

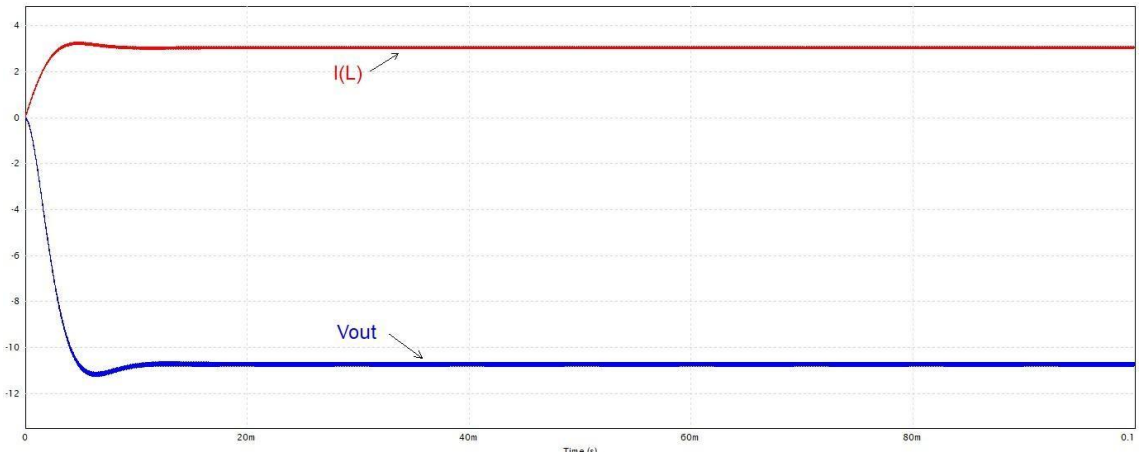


Figure 2 Step-Down Operation

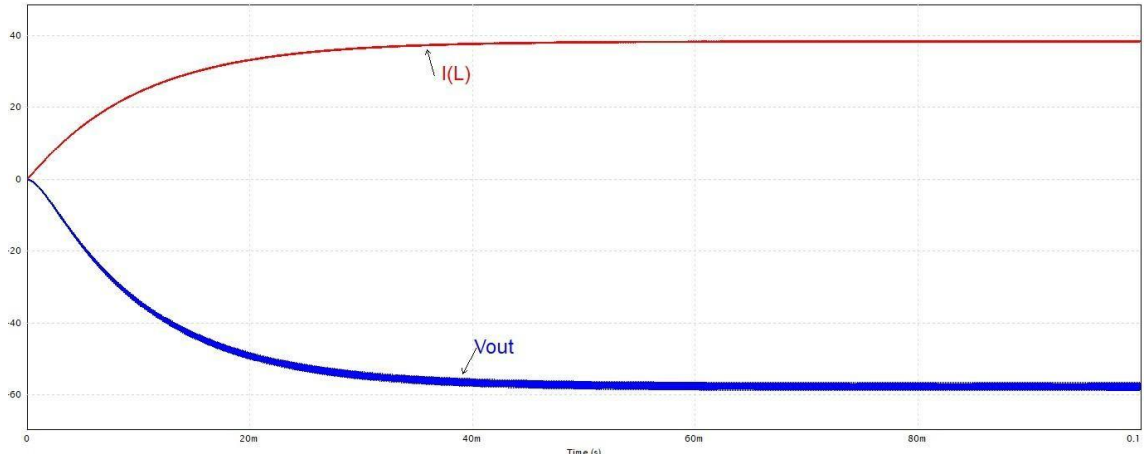


Figure 3 Step-Up Operation

Parameter Sensitivity

Duty Cycle
- Determines both magnitude and direction of voltage conversion.
- For $D < 0.5$, $|V_{out}| < V_{in}$ (buck region).
- For $D > 0.5$, $|V_{out}| > V_{in}$ (boost region).
- Current stress increases rapidly at high duty ratios.

Inductance
- Larger L reduces current ripple in both modes.
- Large L improves damping in boost region but can increase transient overshoot in buck region depending on energy release dynamics.

Capacitance
- Larger C reduces output-voltage ripple.
- Excessive capacitance may prolong settling time during startup.

Summary
- The buck-boost converter provides both step-down and step-up conversion with inverted polarity.
- The same passive network exhibits different transient characteristics in buck and boost regions.
- High duty operation significantly increases inductor current stress.
- A single passive design cannot fully optimize both operating modes in open-loop operation.

This establishes the open-loop buck-boost behavior baseline before implementing voltage feedback and soft-start control.